Network Inference for Hospital Infection Transmission*

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1 Abstract

This study investigates the pathways of inter-ward transmission of Carbapenemase-producing Enterobacteriaceae (CPE) infections by analysing ward-level CPE count data across 3 different hospitals in Victoria, Australia. We employ a dynamic Zero-Inflated Poisson Lognormal (ZIPLN) model to infer the conditional dependence structure between hospital wards, accounting for the temporal dynamics of the count data and the zero-inflation nature of the CPE count data. The results are compared with the Bayesian Mixed Graphical Model (BMGM) to evaluate the necessity of accounting for temporal autocorrelation. The proposed dynamic ZIPLN model demonstrates superior performance in recovering the true conditional dependence structure, particularly in scenarios with strong temporal dependence and high-dimensional settings. The findings provide actionable insights for health policy and hospital infection control strategies.

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2 Introduction

2.1 Motivation and Research aims

Carbapenemase-producing Enterobacteriaceae (CPE) represent a critical public health threat globally and locally, associated with substantial patient morbidity, elevated mortality, and significant financial burdens on healthcare systems (Murray et al. (2022)). In Victoria, Australia, despite the implementation of a centralized "search and contain" policy, CPE cases have risen steadily in recent years (Lane et al. (2021)). Primary acquisition remains strongly linked to international travel and exposure to overseas healthcare facilities (Lane et al. (2021)).

The Victorian CPE guidelines, first released in December 2015, recommend that patients colonised with CPE be managed under contact precautions, including isolation in single rooms, the use of personal protective equipment (PPE), and enhanced environmental cleaning. In addition, screening is recommended for patients with a history of overseas travel or healthcare exposure within the previous 12 months upon hospital admission (Department of Health, Victoria (2026)), aligning with international practices (Solter et al. (2018)).

Despite these proactive guidelines, a fundamental structural vulnerability persists within the Victorian health system: the lack of a centralized real-time laboratory tracking system or alert mechanism to monitor CPE-colonised individuals across healthcare sites (Department of Health, Victoria (2026)). As a result, colonised patients may be transferred between hospitals or readmitted after discharge without sharing their CPE status, limiting the implementation of appropriate infection control measures (Donker et al. (2010); Solter et al. (2018); Lee et al. (2016)). Outbreaks of CPE infections are believed to be driven out of intensive care units (ICUs) and high-risk wards, with subsequent transmission to other wards and hospitals through patient transfers. However,

the specific pathways of inter-ward transmission remain poorly understood, hindering the development of effective infection control strategies. Up to now, there has been no systematic analysis of the inter-ward transmission pathways of CPE infections in Victoria, Australia.

We aim to investigate the pathways of inter-ward transmission of CPE infections by analysing ward-level CPE count data across 3 different campuses from a single healthcare provider in Victoria, Australia. Our goal is to clarify the conditional dependence structure of tranmission between hospital wards to identify the key hubs of healthcare-associated CPE transmission. To achieve this, we employ the Gaussian graphical model (GGMs) to infer the conditional dependence structure between wards, while accounting for the temporal dynamics of the count data and the zero-inflation nature of the CPE count data.

Gaussian graphical models (GGMs) are a convenient framework to describe the conditional dependence structure between random variables. In practice, the GGMs requires the assumption of multivariate normality, which is hard to satisfy in many real-world applications, particularly count data. And it is more challenging when the count data are sparse and zero-inflated, which is the case for hospital infection data. For multivariate count data, the Poisson Lognormal (PLN) framework provides an alternative by modelling count observations through latent Gaussian variables, allowing dependence between variables to be represented through the covariance or precision structure of the latent variables (Aitchison and Ho (1989); Chiquet et al. (2019)).

Nonetheless, hospital infection datasets present two distinct modeling hurdles: extreme zero-inflation and temporal autocorrelation. While univariate Zero-Inflated Poisson (ZIP) models (Lambert (1992)) and multivariate extensions (MZIP, ZINB) accommodate structural zeros, their parameterization of inter-variable correlation remains constrained. Bayesian multivariate zero-inflated frameworks (Lee et al. (2020)) and mixed graphical

models (Florez et al. (2025)) have progressed the joint modeling of dependent zero-inflated counts. More recently, Batardère et al. (2025) introduced the Zero-Inflated Poisson Lognormal (ZIPLN) framework, which leverages a latent Gaussian layer to explicitly decouple structural zero-inflation mechanisms from the underlying Poisson-lognormal dependencies.

Beyond spatial and structural zeros, longitudinal surveillance data inherently exhibit temporal dynamics influenced by dynamic patient transfers, shifting infection control policies, and seasonality. Failing to account for temporal dependence risks identifying spurious correlations between wards.

To address these challenges, bridging the ZIPLN framework with temporal dynamics, we propose a dynamic ZIPLN state-space model. By embedding autoregressive latent variables into the latent layer, our model simultaneously accounts for zero-inflation, multivariate spatial dependencies across hospital wards, and temporal autocorrelation over time. Applying this extended framework to multi-center Victorian ward-level data enables us to infer the true conditional dependence network of CPE transmission, providing actionable insights for health policy and hospital infection control strategies.

2.2 Notation

Let $Y_{t,j}$ denote the observed count of carbapenemase-producing Enterobacterales (CPE) cases in ward j ($j = 1, \dots, P$) during month t ($t = 1, \dots, T$). The observed monthly count data have two key characteristics: (i) non-negative integer values, and (ii) a large proportion of zero counts.

Note that, some months contain no recorded cases across the observed wards and are therefore not represented in the raw dataset. Therefore, the time index t is defined as

the ordered sequence of months with at least one observed CPE case across the wards. The total number of months with at least one observed CPE case is denoted as T.

The conditional dependence structure between wards is represented by a graph $G = (V, E)$, where $V = \{1, \ldots, P\}$ is the set of nodes corresponding to the wards and E is the set of edges representing conditional dependencies between wards. The goal of this study is to estimate the graph G from the observed count data $\{Y_{t,j}\}$ while accounting for zero-inflation and temporal dynamics.

2.3 Literature Review

2.3.1 Bayesian Mixed Graphical Models (BMGM)

To address the challenges, Roy and Dunson (2020) introduced a class of pairwise graphical models for count data, which enables the estimation of both positive and negative conditional dependencies through conditionally specified distributions. This framework was later extended to accommodate zero-inflated mixed data types by Florez et al. (2025). The Bayesian Mixed Graphical Model (BMGM) allows for the joint modeling of zero-inflated counts and continuous variables, providing a flexible framework for estimating conditional dependencies in complex datasets. This framework has been applied to various domains, including genomics, epidemiology, and social network analysis, where zero-inflated count data are prevalent.

$$f(Y_1, \ldots, Y_P) \propto \exp\left(\sum_{j=1}^{P} \theta_j(Y_j) + \sum_{j<k} \Omega_{j,k} g_j(Y_j) g_k(Y_k)\right)$$

where $\theta_j(Y_j)$ is the node potential function for variable Y_j, $\Omega_{j,k}$ is the edge potential function representing the conditional dependence between variables Y_j and Y_k, and $g_j(Y_j)$ is a transformation function that maps the observed count data to a suitable scale

for modeling. The BMGM framework allows for the estimation of both positive and negative conditional dependencies, providing a comprehensive view of the underlying network structure.

The limitation of BMGM is that it assumes independence across temporal observations, which may lead to spurious associations between wards when temporal dependence is present. In the context of hospital infection data, where patient transfers and infection control measures can induce temporal autocorrelation, failing to account for temporal dynamics can result in misleading estimates of the conditional dependence structure. However, it is important to note that BMGM provides a useful benchmark for evaluating the performance of our proposed dynamic ZIPLN model, especially in scenarios where temporal dependence is weak or absent.

2.3.2 Zero-Inflated Poisson Log-Normal (ZIPLN) Model

Following Batardière et al. (2025), excess zeros in spatial count data can be modeled using a Zero-Inflated Poisson Log-Normal (ZIPLN) framework. In their standard ZIPLN model, observed count are decomposed into two latent processes: a Bernoulli process $W_{t,j} \sim Bernoulli(\pi_{t,j})$ that capture the structural zeros, and a latent Gaussian vector $H_t = (H_{t,1}, \dots, H_{t,P})^T \sim N(\mu_t, \Sigma_H)$ driving the Poisson intensity of the count data. The observed count $Y_{t,j}$ is then generated as follows:

$$Y_{t,j} = \begin{cases} 0, & \text{if } W_{t,j} = 0, \\ Poisson(exp(H_{t,j})), & \text{if } W_{t,j} = 1. \end{cases}$$

While the static ZIPLN formulation effectively captures overdispersion and static spatial co-occurrence, it assumes temporal independence across time points (t). In reality, the count data of CPE cases in different wards are collected over time, and the temporal

dynamics can be influenced by various factors, such as patient movement between wards, infection control measures, and seasonal variations. Therefore, it is important to account for the temporal dynamics in the analysis to obtain accurate estimates of the conditional dependence structure between hospital wards.

Using the same approach as Batardière et al. (2025), we extend the ZIPLN framework to a dynamic setting by introducing autoregressive latent processes for both the Poisson intensity and zero-inflation components. This allows us to capture the temporal autocorrelation in the count data while still accounting for the zero-inflation and overdispersion characteristics of the data.

3 METHODOLOGY

Under the Zero-Inflated Poisson framework, the marginal distribution of the observed count $Y_{t,j}$ is a mixture of a point mass at zero and a Poisson distribution, given by:

$$Y_{t,j} \sim \text{ZIP}(\lambda_{t,j}, \pi_{t,j}),$$

with probability mass function:

$$P(Y_{t,j} = y_{t,j}) = \begin{cases} \pi_{t,j} + (1 - \pi_{t,j})e^{-\lambda_{t,j}}, & \text{if } y_{t,j} = 0, \\ (1 - \pi_{t,j})\frac{e^{-\lambda_{t,j}}\lambda_{t,j}^{y_{t,j}}}{y_{t,j}!}, & \text{if } y_{t,j} > 0. \end{cases}$$

where $\lambda_{t,j} > 0$ represents the expected Poisson intensity and $\pi_{t,j} \in (0, 1)$ denotes the probability of a structural zero for ward j at month t.

To account for temporal dynamics and estimate the inter-ward conditional dependence network, we propose a dynamic ZIPLN state-space model that incorporates latent

autoregressive processes for both the Poisson intensity and zero-inflation components. This allows us to capture the temporal autocorrelation in the count data while still accounting for the zero-inflation and overdispersion characteristics of the data. The model has two stages: the dynamic ZIPLN state-space model for the observed counts and a post-hoc Gaussian Graphical Model (GGM) for network inference based on the latent innovations.

3.1 DYNAMIC ZIPLN STATE-SPACE MODEL

To explicitly capture stochastic temporal autocorrelation rather than imposing a strict linear trend, we formulate a dynamic ZIPLN state-space model. Here, the time-varying intensity $\lambda_{t,j}$ and zero-inflation probability $\pi_{t,j}$ are linked to latent autoregressive processes:

For each ward (j) and time point (t), instead of assuming independence, we model the Poisson-intensity and zero-inflation components as follows:

$$\log(\lambda_{t,j}) = \beta_{0,j} + S_{t,j},$$
$$logit(\pi_{t,j}) = Z_{t,j},$$

where $\beta_{0,j}$ represents the baseline log-intensity for ward j, $S_{t,j}$ is a latent intensity process, and $Z_{t,j}$ is a latent zero-inflation process.

3.1.1 Zero-Inflation Latent Process (Z)

The zero-inflation logit probability $\pi_{t,j}$ is modeled by a ward-specific latent process $Z_{t,j}$, which follows an independent stationarised autoregressive process of order 1 (AR(1)):

$$Z_{1,j} \sim \text{Normal}\left(c_{zj}, \frac{\sigma_{zj}}{\sqrt{1-\rho_j^2}}\right), \quad \text{for } t = 1$$

$$Z_{t,j} \mid Z_{t-1,j} \sim \text{Normal}\left((1-\rho_j)c_{zj} + \rho_j Z_{t-1,j}, \sigma_{zj}^2\right), \quad \text{for } t = 2, \ldots, T$$

where c_{zj} is the drift parameter, $\rho_j \in (-1, 1)$ is the ward-specific AR(1) coefficient, and σ_{zj}^2 is the innovation variance of the zero-inflation latent process.

3.1.2 Poisson-Intensity Latent Process (S)

Similarly, the Poisson-intensity latent process $S_{t,j}$ for each ward, which captures temporal dynamics in true ward-level infection risk, follows an independent stationary AR(1) process:

$$S_{1,j} \sim \text{Normal}\left(c_{sj}, \frac{\sigma_{sj}}{\sqrt{1-\phi_j^2}}\right), \quad \text{for } t = 1$$

$$S_{t,j} \mid S_{t-1,j} \sim \text{Normal}\left((1-\phi_j)c_{sj} + \phi_j S_{t-1,j}, \sigma_{sj}^2\right), \quad \text{for } t = 2, \ldots, T$$

where c_{sj} is the stationary mean, $\phi_j \in (-1, 1)$ is the ward-specific AR(1) coefficient, and σ_{sj}^2 is the innovation variance of the Poisson-intensity latent process.

3.2 Post-Hoc Gaussian Graphical Model (GGM) Network Inference

Standard correlation measures applied directly to raw count data $Y_{t,j}$ conflate temporal autocorrelation, Poisson sampling variance, and structural zero-inflation. To isolate true contemporaneous inter-ward transmission pathways, we construct a Gaussian Graphical Model on the latent dynamic innovations.

3.2.1 Extraction of Latent Innovations

For $t = 2, \dots, T$, we define the latent log-intensity innovation vector $\mathbf{r}_t = (r_{t,1}, \dots, r_{t,P})' \in \mathbb{R}^P$ as the white-noise residual remaining after subtracting the AR(1) temporal expectation:

$$r_{t,j} = S_{t,j} - \mathbb{E}[S_{t,j} \mid S_{t-1,j}] = S_{t,j} - ((1 - \phi_j)c_{sj} + \phi_j S_{t-1,j}),$$

3.2.2 Precision Matrix Estimation & Network Identification

We standardize the innovation matrix $\widehat{\mathbf{R}}$ and assume the standardized innovation vectors follow a zero-mean multivariate normal distribution:

$$\hat{\mathbf{r}}_t \sim \mathcal{N}_P(\mathbf{0},\ \Sigma), \quad t = 2, \dots, T$$

The conditional dependence graph $G = (V, E)$ across the set of hospital wards $V = \{1, \dots, P\}$ is uniquely determined by the sparse precision matrix $\Omega = \Sigma^{-1}$:

$$\Omega_{j,k} = 0 \quad \iff \quad \hat{r}_{t,j} \perp \hat{r}_{t,k} \mid \hat{\mathbf{r}}_{t,\setminus\{j,k\}}$$

Off-diagonal elements $\Omega_{j,k} \neq 0$ represent direct conditional dependencies between wards j and k, indicating potential pathways of CPE transmission.

We estimate the sparse graph topology $\widehat{G}$ and precision matrix $\widehat{\Omega}$ using Bayesian spike-and-slab MCMC graph selection (modelSelectionGGM) (Sulem et al. (2025)). Partial correlation coefficients $P_{j,k} \in [-1, 1]$ indicating the normalized strength of direct ward-to-ward transmission are calculated as:

$$P_{j,k} = -\frac{\Omega_{j,k}}{\sqrt{\Omega_{j,j}\Omega_{k,k}}}, \quad \text{for } j \neq k$$

4 PRIOR SPECIFICATION AND POSTERIOR INFERENCE

4.1 PRIOR SPECIFICATION

To complete the Bayesian framework, we assign weakly informative prior distributions to all parameters. These priors provide light regularization to stabilize parameter estimation and avoid overfitting in sparse count settings, while allowing the observed data to primary drive posterior inference.

For each ward $j = 1, \ldots, P$, the baseline intercept and stationary means are assigned Gaussian priors centered at zero:

$$\begin{aligned} \mathrm{B}_{0,j} &\sim N(0,1), \\ c_{Z,j} \sim N(0, 0.1), \quad c_{S,j} &\sim N(0, 0.1), \end{aligned}$$

To enforce stationarity on the latent AR(1) processes, the autocorrelation parameters ρ_j and ϕ_j are transformed to the $(-1, 1)$ domain and assigned Beta priors shifted and scaled to $(-1, 1)$, or equivalently:

$$\frac{\rho_j + 1}{2} \sim \mathrm{Beta}(a_\rho, b_\rho), \quad \frac{\phi_j + 1}{2} \sim \mathrm{Beta}(a_\phi, b_\phi),$$

with $a_\rho = a_\phi = 2$ and $b_\rho = b_\phi = 2$ to place light probability density away from extreme boundaries (± 1).

The innovation standard deviations $\sigma_{Z,j}$ and $\sigma_{S,j}$ governing the latent process variations are assigned Half-Normal prior distributions:

$$\sigma_{Z,j} \sim \text{Half-Normal}(0,1), \quad \sigma_{S,j} \sim \text{Half-Normal}(0,1).$$

4.2 POSTERIOR INFERENCE

Let

$$\theta = \left(\beta_0, c_Z, c_S, \rho, \phi, \sigma_Z, \sigma_S\right)$$

denote the full set of model parameters, and let Z and S represent the matrices of latent zero-inflation and Poisson-intensity states across all time points $t = 1, \dots, T$ and wards $j = 1, \dots, P$.

By Bayes' Theorem, the joint posterior distribution of the parameters and latent state vectors given the observed counts Y is proportional to the product of the marginal ZIP likelihood, the latent state transition densities, and the parameter priors:

$$\begin{aligned} p(\beta_{0j}, c_{zj}, c_{sj}, \rho_j, \phi_j, \sigma_{zj}^2, \sigma_{sj}^2 \mid Y) \propto\ & p(Y \mid \beta_{0j}, c_{zj}, c_{sj}, \rho_j, \phi_j, \sigma_{zj}^2, \sigma_{sj}^2) \\ & \times p(\beta_{0j}) p(c_{zj}) p(c_{sj}) p(\rho_j) p(\phi_j) p(\sigma_{zj}^2) p(\sigma_{sj}^2) \end{aligned}$$

4.3 MARKOV CHAIN MONTE CARLO (MCMC) SAMPLING VIA STAN

Because the joint posterior distribution of high-dimensional dynamic models cannot be solved analytically, Bayesian inference is performed using Markov Chain Monte Carlo (MCMC) methods to generate random draws from the target posterior distribution.

Traditional MCMC algorithms, such as Metropolis-Hastings and Gibbs sampling, explore the parameter space by taking small random steps or updating parameters conditionally one at a time. However, with the high dimensional or heavily correlated parameter spaces, these methods can be inefficient, leading to slow convergence and poor mixing of the Markov chains. In that case, the strong posterior correlation between latent dynamic process ($Z_{t,j}$ and $S_{t,j}$) and structural parameter (θ) can lead to slow convergence and poor mixing of the Markov chains.

To overcome these limitations, we perform inference using Stan, which implements Hamiltonian Monte Carlo (HMC) alongside the No-U-Turn Sampler (NUTS). Unlike traditional MCMC, HMC leverages gradient information to propose new parameter values, allowing for larger and more informed jumps in the parameter space. This dramatically reduces autocorrelation, accelerates convergence, and ensures robust exploration of complex target distributions making it particularly well-suited for our zero-inflated time-series model.

5 SIMULATION

We conducted simulation studies to evaluate the ability of the proposed dynamic zero-inflated Poisson log-normal (ZIPLN) model to recover the conditional dependence structure between wards under different combinations of dimensionality and temporal dependence. The simulations were designed to assess network recovery under data characteristics representative of the CPE surveillance data, including a high proportion of zero counts, overdispersion, and temporal dependence.

5.1 Simulation Design and Data Generation

For each simulation, count observations $Y_{t,j}$ were generated for time points $t = 1, \dots, T$ and wards $j = 1, \dots, P$ from a zero-inflated Poisson observation model. The Poisson log-intensity was determined by a latent process (S_{t,j}), while a separate latent process (Z_{t,j}) governed the probability of structural zeros. Specifically,

$$Y_{t,j} \sim \text{ZIP}(\lambda_{t,j}, \pi_{t,j}), \quad \text{with } \lambda_{t,j} = \beta_{0,j} + S_{t,j}, \quad \pi_{t,j} = \text{logit}^{-1}(Z_{t,j}),$$

The latent processes $(S_t = (S_{t,1}, \dots, S_{t,P})^\top)$ and $(Z_t = (Z_{t,1}, \dots, Z_{t,P})^\top)$ were generated independently. Each process followed a stationary AR(1) structure. Dependence between wards was introduced through the covariance matrix of the innovations in the (S_t) process. Let $\epsilon_t = (\epsilon_{t,1}, \dots, \epsilon_{t,P})^\top \sim N_P(0, \Sigma)$ and precision matrix $\Omega = \Sigma^{-1}$, where the off-diagonal elements of Ω encode the conditional dependence between wards.

We considered three simulation scenarios. The scenarios varied the number of wards, the number of time points, and the strength of temporal dependence, while retaining a common underlying network structure where appropriate. The simulation settings are summarised below.

5.1.1 Scenario 1: Static Spatial Dependence with No Temporal Autocorrelation

In scenario 1, we consider a static spatial baseline with $P = 10$ wards observed over $T = 500$ time points. The precision matrix Ω was initialized as an identity matrix with two off-diagonal correlated pairs ($\Omega_{1,5} = \Omega_{5,1} = -0.8$ and $\Omega_{2,3} = \Omega_{3,2} = 0.8$). Temporal autocorrelation and mean shifts were eliminated by setting the autoregressive parameters and intercepts to zero ($\rho = 0$, $\phi = 0$, $c_{Z,j} = 0$, $c_{S,j} = 0$), alongside baseline log-intensity $\beta_0 = 0$ and innovation noise standard deviation $\sigma_Z = 0.1$

5.1.2 Scenario 2: Static Spatial Dependence with Temporal Autocorrelation

Scenario 2 used the same number of wards, time points, and underlying network as Scenario 1, but introduced strong temporal dependence. The AR coefficients were set to ($\rho = 0.8$) for the zero-inflation process and ($\phi = 0.8$) for the log-intensity process. The corresponding process intercepts were set to ($c_{Z,j} = 0.5$) and ($c_{S,j} = 1$), with ($\beta_{0,j} = 0$) and ($\sigma_Z = 0.1$).

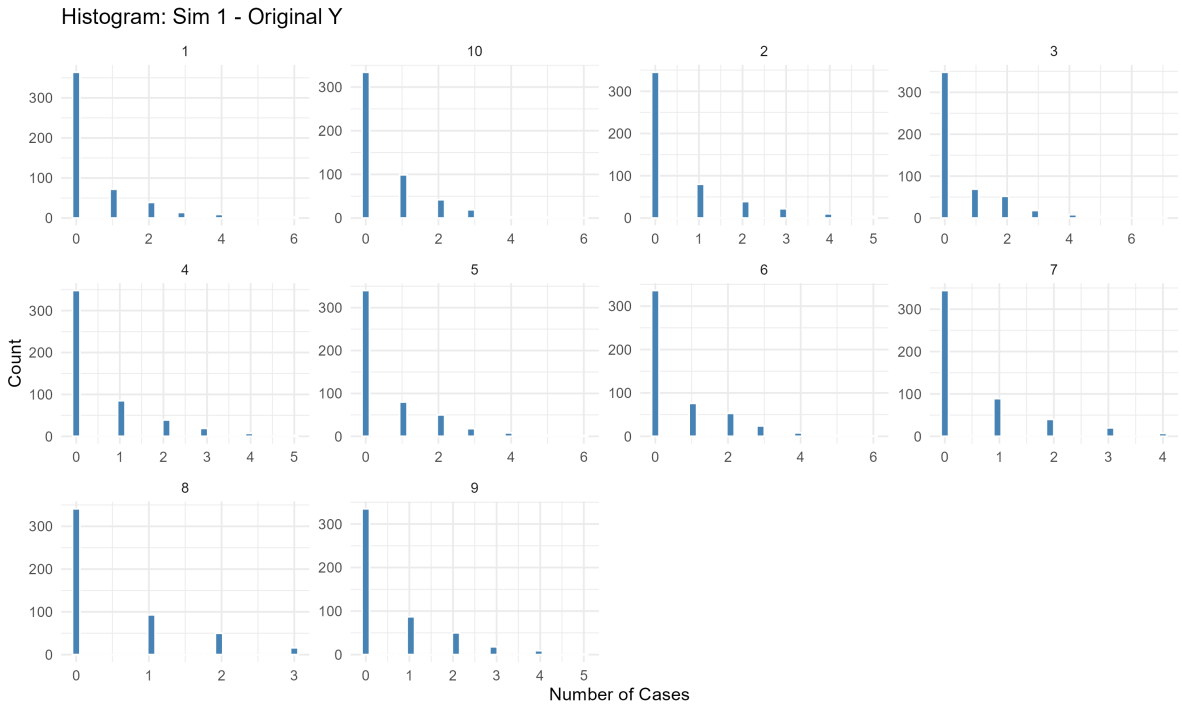
5.1.3 Scenario 3: High-Dimensional Setting

Scenario 3 investigated a higher-dimensional setting with (P=30) wards and only (T=100) time points. The precision matrix contained ten designated off-diagonal edge pairs. The latent processes otherwise followed the same temporal specification as Scenario 2. This scenario was designed to examine network recovery when the number of variables substantially exceeds the number of temporal observations.

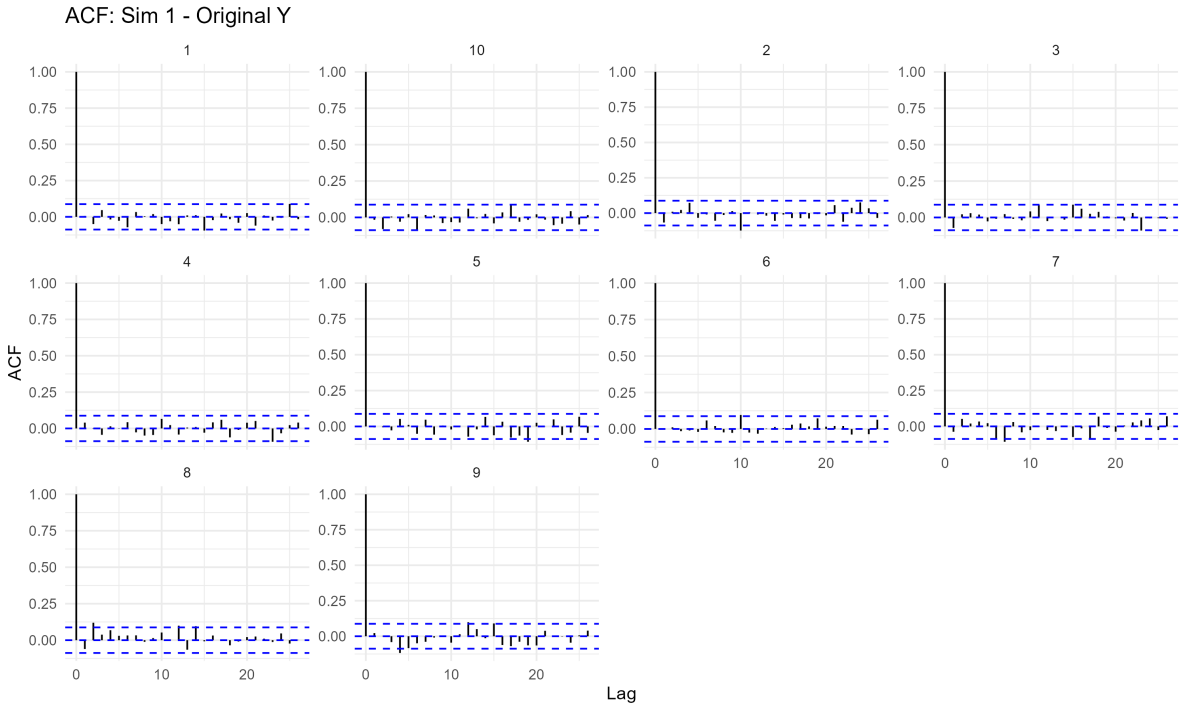
5.2 Exploratory Data & Autocorrelation Characteristics

Figure 1 and Figure 2 present the empirical distributions and autocorrelation functions (ACFs) of the simulated count observations. Scenarios 1 and 2 exhibit similar marginal count distributions, characterised by a large proportion of zero observations and a right-skewed distribution of positive counts. This behaviour reflects the zero-inflated observation mechanism combined with heterogeneity in the latent log-intensity process.

It is hard to see the difference in temporal autocorrelation between Scenarios 1 and 2. But if we look at deeply the ACF plots, we can see that the temporal autocorrelation is more pronounced in Scenario 2 than in Scenario 1. As expected, the ACFs in Scenario 1 show little evidence of temporal autocorrelation because both latent AR coefficients are

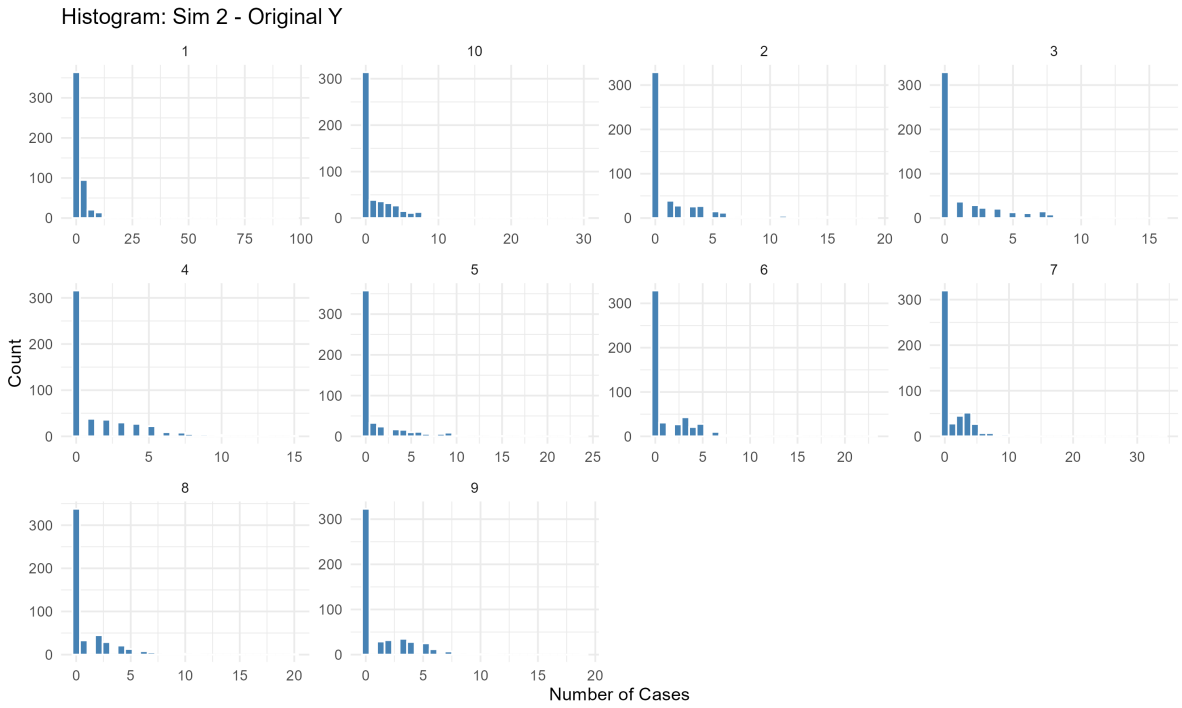


(a) Scenario 1: Histogram

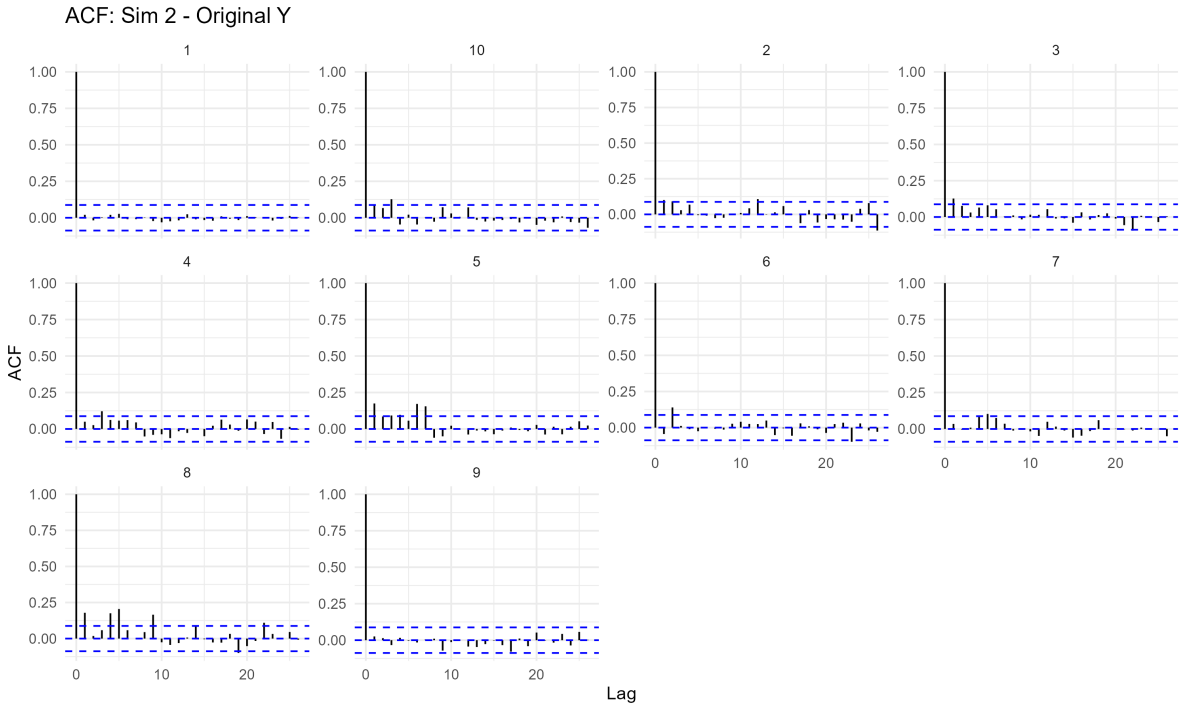


(b) Scenario 1: ACF

Figure 1: The empirical distributions and autocorrelation functions (ACFs) of the simulated counts in Scenarios 1



(a) Scenario 2: Histogram



(b) Scenario 2: ACF

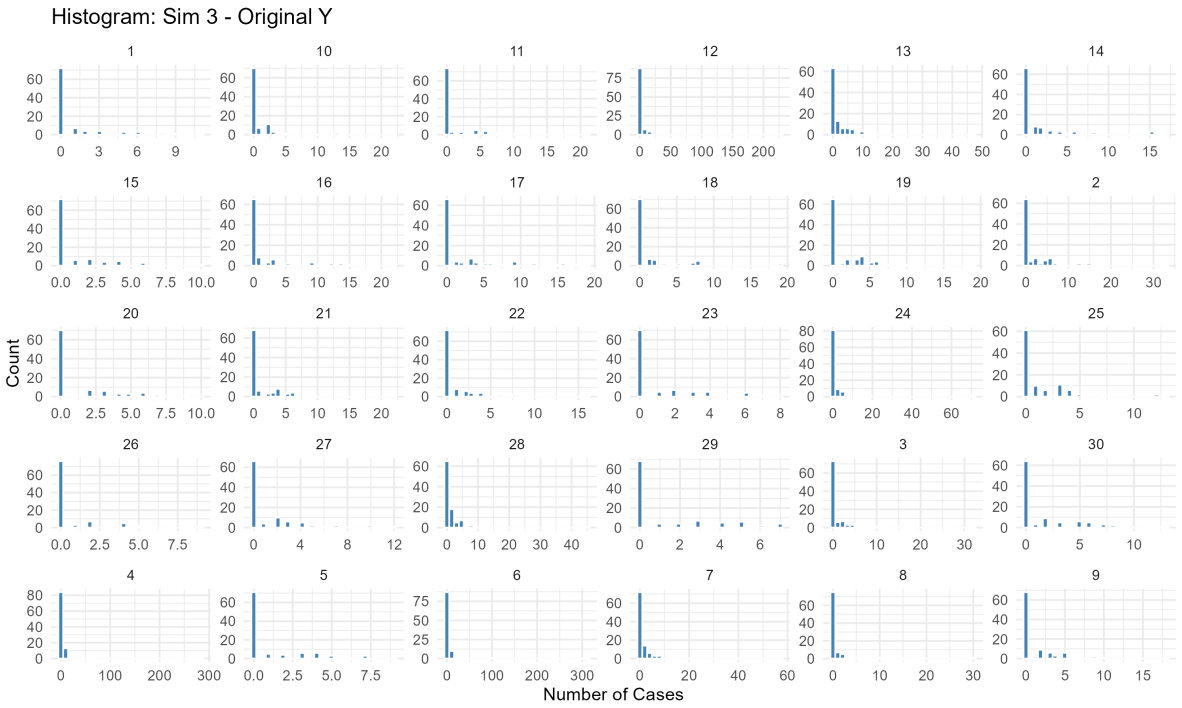
Figure 2: The empirical distributions and autocorrelation functions (ACFs) of the simulated counts in Scenarios 2

set to zero. In contrast, Scenario 2 exhibits pronounced temporal dependence, consistent with the AR coefficients of ($\rho = \phi = 0.8$) used to generate both latent processes.

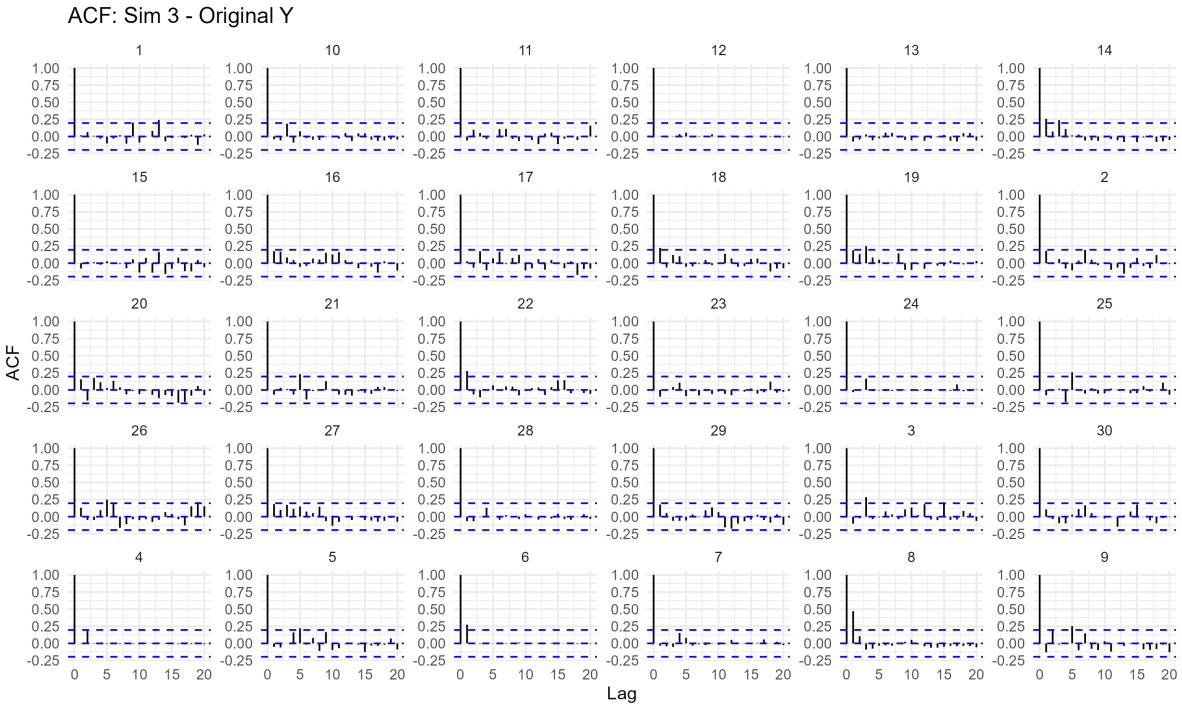
Scenario 3, with a larger number of wards and fewer time points, retains the strong temporal dependence of Scenario 2 but contains substantially more wards and fewer time points. Consequently, we expect that the empirical ACF estimates are more variable and provide less information about the underlying temporal dynamics. The combination of (P=30) and (T=100) also creates a substantially more challenging network estimation problem. And Figure 3 illustrates the empirical distribution, ACFs of the simulated counts in Scenario 3. The histogram plot shows a similar zero-inflated and right-skewed distribution of positive counts as in Scenarios 1 and 2. The ACF plot also exhibits greater variability due to the limited number of time points relative to the number of wards, making it more difficult to discern the underlying temporal structure.

Table 1 summarises the key metrics of the simulated counts across the three scenarios. As expected, when there is no temporal dependence (Scenario 1), all parameters are recovered accurately, nearly unbiased across β_0, ρ, c_Z, c_S. Every parameter is able to coverage the true value within the 95% credible interval. The RMSE for the intercepts in this case is 0.098, which is very small and the wider CIs for ρ and ϕ ($[-0.784, 0.905]$) are expected when sampling latent AR coefficients near zero with weak priors.

For Scenarios 2 and 3, where strong temporal dependence is present ($\phi = \rho = 0.8$), the model successfully recovers total baseline log-intensity ($\beta_0^* = \beta_0 + c_S$) and infection persistence (ϕ), while exhibiting expected conservative shrinkage for zero-inflation persistence (ρ). Although individual estimates for β_0 and c_S reflect their additive location identity ($\mathbb{E}[\log \lambda_{t,j}] = \beta_{0,j} + c_{S,j}$), the total baseline mean is recovered with near-zero bias ($\hat{\beta}_0^* = 0.966$ vs True 1.000 in Scenario 2), as the tight prior on c_S cleanly transfers spatial location into β_0. The primary infection persistence ϕ is recovered with high precision across both settings ($\hat{\phi} = 0.775$ for $P = 10, T = 500$; $\hat{\phi} = 0.739$ for



(a) Scenario 3: Histogram (P=30, T=100)



(b) Scenario 3: ACF (P=30, T=100)

Figure 3: The empirical distributions and autocorrelation functions (ACFs) of the simulated counts in Scenario 3

Table 1: The summary metrics of the simulated counts in 3 scenarios

Parameter Recovery Performance Across Simulation Scenarios

Posterior Mean, Bias, Root Mean Squared Error (RMSE), and 95% Credible Intervals

Parameter	True Value	Posterior Mean	Bias	RMSE	95% CI
Scenario 1					
β_0	0.000	0.019	0.019	0.179	[−0.284, 0.305]
ρ	0.000	0.032	0.032	0.497	[−0.784, 0.905]
φ	0.000	0.120	0.120	0.504	[−0.741, 0.871]
c_Z	0.000	0.013	0.013	0.098	[−0.170, 0.190]
c_S	0.000	0.001	0.001	0.098	[−0.191, 0.195]
Scenario 2					
β_0	0.000	0.956	0.956	0.975	[0.622, 1.281]
ρ	0.800	0.417	−0.383	0.702	[−0.673, 0.996]
φ	0.800	0.775	−0.025	0.119	[0.578, 0.900]
c_Z	0.500	0.154	−0.346	0.364	[−0.063, 0.328]
c_S	1.000	0.010	−0.990	0.995	[−0.181, 0.200]
Scenario 3					
β_0	0.000	0.888	0.888	0.953	[0.343, 1.366]
ρ	0.800	0.516	−0.284	0.623	[−0.567, 0.997]
φ	0.800	0.739	−0.061	0.268	[0.429, 0.924]
c_Z	0.500	0.098	−0.402	0.415	[−0.104, 0.289]
c_S	1.000	0.009	−0.991	0.996	[−0.186, 0.203]

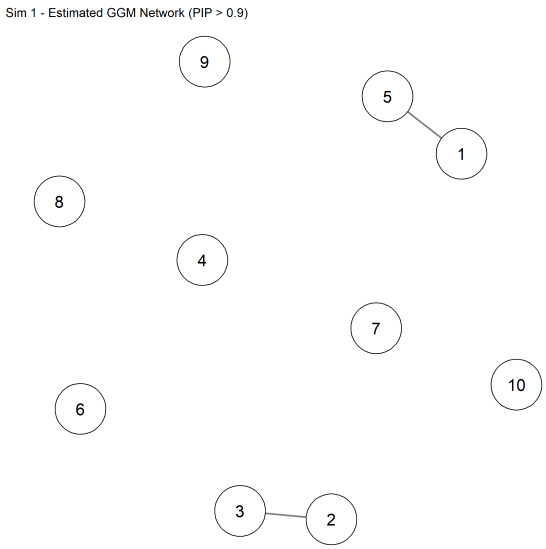
$P = 30, T = 100$), ensuring within-ward autocorrelation is accurately stripped away to yield pure innovation residuals $\hat{\mathbf{r}}_t$ for network selection. Lastly, the attenuation in zero-inflation persistence ($\hat{\rho} \approx 0.420 - 0.516$) reflects expected prior regularization under sparse binary signals, where structural and sampling zeros are observationally confounded; crucially, the zero-inflation layer absorbs these excess zeros, shielding ϕ from distortion.

5.2.1 Comparative Network Recovery Performance

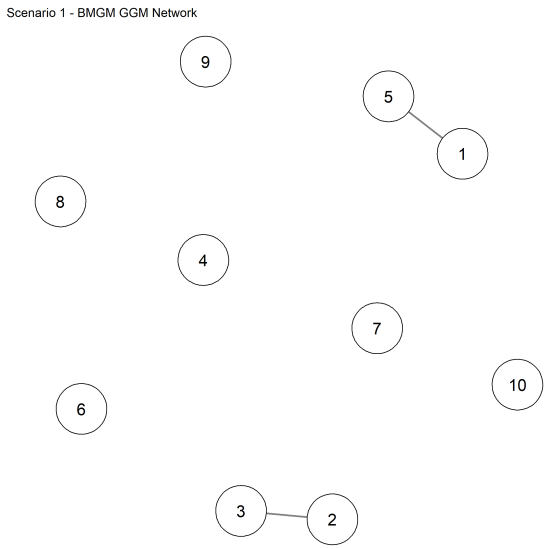
To evaluate the necessity of accounting for temporal autocorrelation, we benchmarked the dynamic ZIPLN model against the Bayesian Mixed Graphical Model (BMGM; Florez et al. (2025)). While BMGM accommodates zero-inflated mixed data types, it assumes independence across temporal observations.

Figure 4 compares the true and estimated networks obtained from the proposed dynamic ZIPLN model and the Bayesian Mixed Graphical Model (BMGM) across Scenarios 1 ($\rho = \phi = 0.0$). In Scenario 1, where temporal dependence was absent, the BMGM (with the PPI threshold equal to 0.8) produced a network broadly similar to that obtained from the dynamic ZIPLN model. This is expected because the data-generating process in this scenario does not contain temporal dependence that needs to be accounted for.

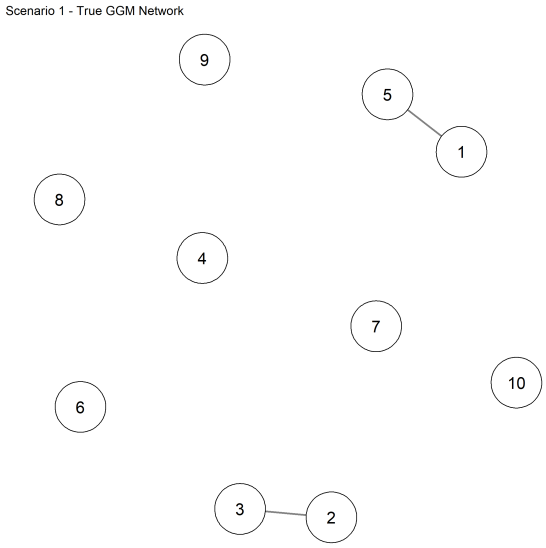
On the other hand, Figure 5 show the comparision between the true, estimated networks obtained from the proposed dynamic ZIPLN and BMGM models in Scenario 2. Due to the presence of strong temporal dependence, the BMGM produced a network that differed substantially from the true network. It only identified 1 true positive edge out of 2, and required the PPI threshold more relaxed, which is equal to 0.5, suggesting that the BMGM model started to misattribute temporal dependence to conditional dependence between wards. In contrast, the dynamic ZIPLN model was able to recover the principal edges. This comparison illustrates the potential effect of unmodelled



(a) ZIPLN model on simulated data scenario 1



(b) Bayesian Mixed Graphical Model (BMGM) on simulated data scenario 1

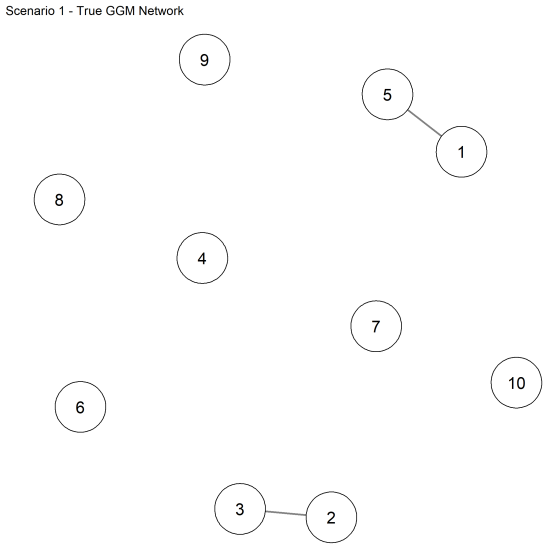


(c) True network

Figure 4: The true and estimated networks obtained from the proposed dynamic ZIPLN and BMGM models across Scenarios 1



(a) ZIPLN model on simulated data scenario 2(b) Bayesian Mixed Graphical Model (BMGM) on simulated data scenario 2



(c) True network

Figure 5: The true and estimated networks obtained from the proposed dynamic ZIPLN and BMGM models across Scenarios 2

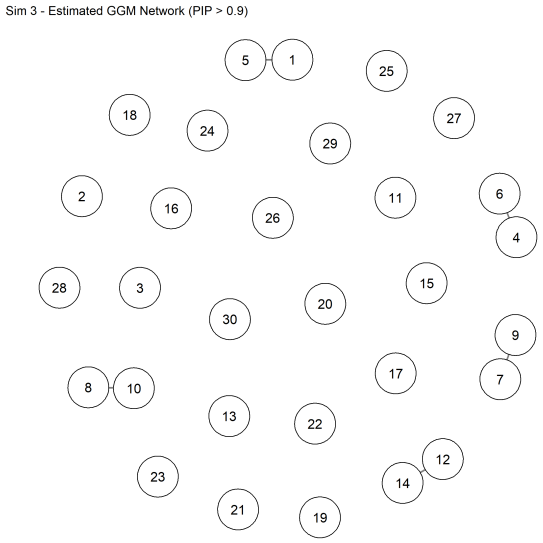
temporal dependence on network estimation: associations induced by persistent temporal processes may be attributed to conditional dependence between wards when temporal structure is ignored.

Figure 6 demonstrates network recovery performance under the challenging high-dimensional setting of Scenario 3 ($P = 30, T = 100$). Across the high number of possible edges, the dynamic ZIPLN model correctly identified 5 true positve edges out of 10, while the BMGM model only identified 2 true positive edges and 1 false positive edge. The deterioration in network recovery performance in Scenario 3 highlights the challenges of estimating conditional dependence structures in high-dimensional settings, particularly when the number of temporal observations is limited relative to the number of wards.

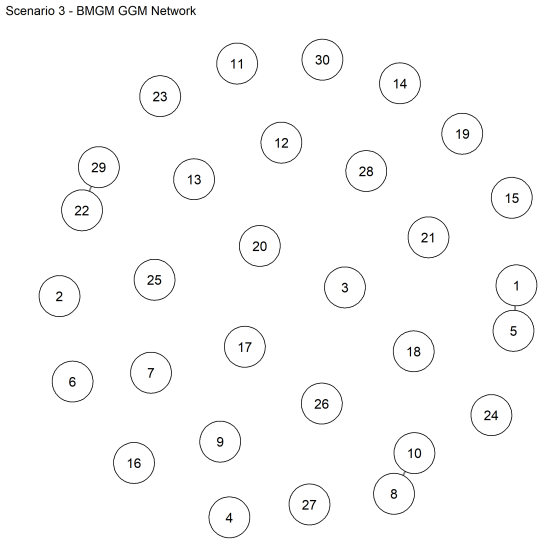
Taken together, these simulations demonstrate that the proposed dynamic ZIPLN framework can recover the underlying conditional dependence structure in low and moderate dimensional settings, including when substantial temporal dependence is present. The results also highlight the importance of explicitly modelling temporal dynamics when estimating conditional dependence from longitudinal count data. The deterioration in Scenario 3 further illustrates the limitations of network recovery when the dimensionality of the system is large relative to the number of temporal observations.

6 Application: Carbapenemase-producing Enterobacteriaceae (CPE) in Victoria, Australia

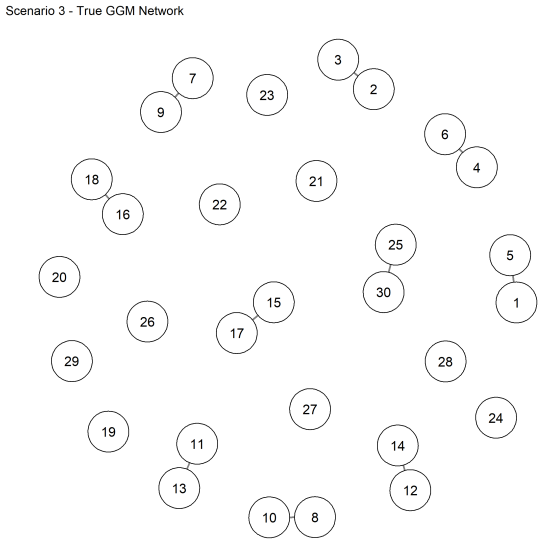
We illustrate the application of the proposed dynamic ZIPLN model to real-world CPE surveillance data collected from three hospitals in Victoria, Australia. The dataset consists of monthly counts of CPE cases across multiple wards from 2013 to 2021. The



(a) ZIPLN model on simulated data scenario 3



(b) Bayesian Mixed Graphical Model (BMGM) on simulated data scenario 3



(c) True network

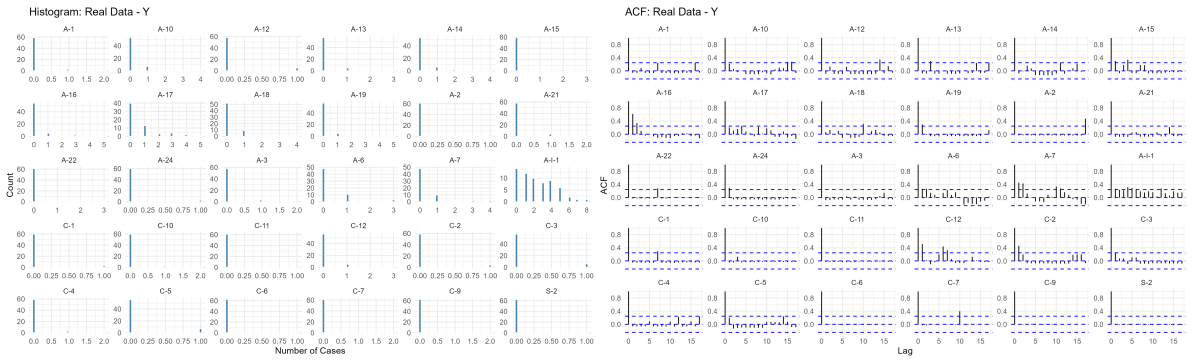
Figure 6: The true and estimated networks obtained from the proposed dynamic ZIPLN and BMGM models across Scenarios 3

data exhibit the characteristic features of zero-inflation, overdispersion, and temporal dependence, making it an ideal candidate for the proposed modeling framework.

6.1 Data overview

The surveillance data were provided by Alfred Health and comprise observations from three hospitals. The dataset contains an encoded ward identifier, which distinguishes individual hospital wards including intensive care units (ICUs) and non-ICU wards, together with the observation month and the corresponding number of CPE cases. This study was approved by the Alfred Health Human Research Ethics Board (Local reference number 445/21).

After preprocessing, the analysis dataset contains 226 observed ward-month combinations across 30 wards and 62 months. Some months contain no recorded cases across the observed wards and are therefore not represented in the raw dataset. The resulting data are highly sparse, with many wards recording zero CPE cases over extended periods. For some wards, more than 80% of the observed counts are zero. Among the non-zero observations, case counts vary considerably across wards and months.



(a) CPE Data - Histogram (b) CPE Data - ACF

Figure 7: The empirical distributions and autocorrelation functions (ACFs) of the CPE data

Table 2: Summary metrics of the CPE data

Victorian CPE Surveillance Data Parameter Estimates Posterior Mean, Standard Deviation, and 95% Credible Intervals across Hospital Wards (P = 60, T = 62)			
Parameter	**Posterior Mean**	**Posterior SD**	**95% CI**
β_0	−0.995	0.707	[−2.234, 0.561]
β_0^* $(\beta_0 + c_S)$	−1.005	0.707	[−2.247, 0.554]
ρ	0.444	0.554	[−0.804, 0.997]
φ	0.462	0.535	[−0.761, 0.997]
c_Z	0.007	0.100	[−0.189, 0.203]
c_S	−0.010	0.100	[−0.206, 0.186]

The observed counts of CPE cases across wards over time are shown in Figure 7. The left panel shows the temporal sparsity and high variance of positive case counts across wards. The right panel displays sample ACFs across wards, confirming the presence of temporal autocorrelation driven by chronic patient colonization, delayed environmental decontamination, and recurring ward-level readmissions.

6.2 Results and Interpretation

6.2.1 Estimated Conditional Dependence Network

Table 2 summarises the estimated parameters of the dynamic ZIPLN model fitted to the CPE data, and Figure 8 presents the estimated conditional dependence network of

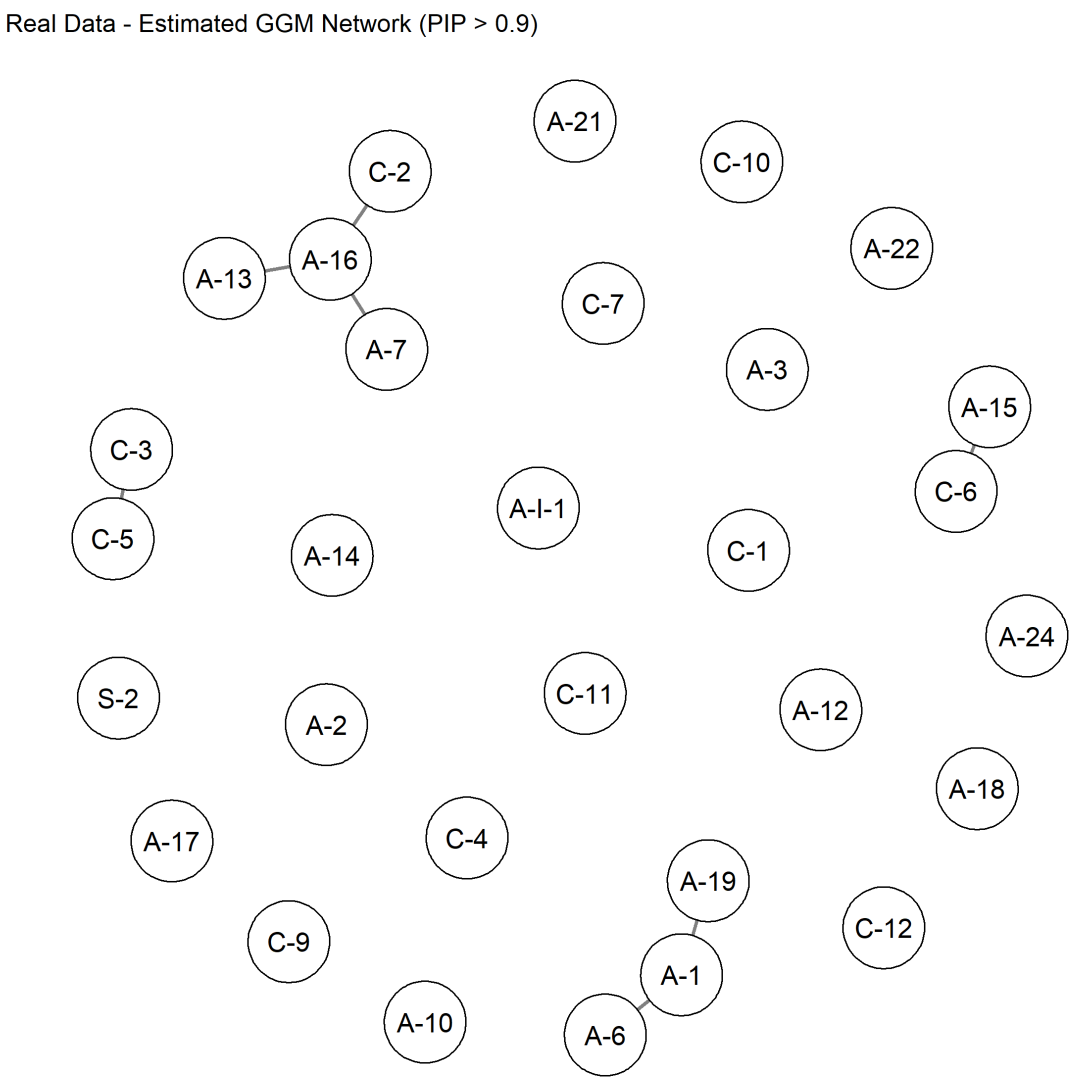


Figure 8: The estimated conditional dependence network of the CPE data

the CPE data. The network contains 30 nodes, each representing a ward, and 5 edges, indicating direct conditional dependencies between wards.

Table 2 summarizes the estimated posterior parameters of the dynamic ZIPLN model fitted to the Victorian CPE surveillance data.

The posterior mean baseline log-intensity across hospital wards was estimated at $\hat{\beta}_0^* = \hat{\beta}_0 + \hat{c}_S = -1.005$ (SD $= 0.707$, 95% CI: $[-2.247, 0.554]$). This corresponds to an expected baseline Poisson intensity of $\exp(-1.005) \approx 0.366$ cases per ward-month across non-zero risk periods. This negative baseline risk reflects the high spatial sparsity of CPE in Victorian health facilities, where individual wards remain free of detected cases for extended periods interrupted by sporadic outbreak clusters. As demonstrated in our simulation studies, the regularizing prior on c_S held process drift near zero ($\hat{c}_S = -0.010, \text{SD} = 0.100$), ensuring that β_0 cleanly absorbed overall ward-level spatial location.

The estimated AR(1) coefficients for the Poisson-intensity latent process ($\hat{\phi} = 0.462, \text{SD} = 0.535$) and zero-inflation latent process ($\hat{\rho} = 0.444, \text{SD} = 0.554$) both indicate moderate positive temporal persistence. Epidemiologically, an autoregressive persistence of ≈ 0.45 confirms that CPE case counts exhibit within-ward temporal memory driven by prolonged length of stay for colonized patients and delayed environmental decontamination. However, because persistence is moderate ($\phi < 0.50$), case series are not endemic or permanent, justifying the model's capacity to filter out temporal memory before Stage 2 network inference. For the posterior standard deviations for the latent innovations, they were small ($\hat{\sigma}_S \approx 0.100$, $\hat{\sigma}_Z \approx 0.100$), confirming that the latent processes fluctuate smoothly around their respective stationary means without absorbing high-frequency cross-ward signals.

Figure 8 presents the sparse conditional dependence graph $\widehat{G} = (V, E)$ inferred from Stage 2 Bayesian spike-and-slab selection on the standardized latent innovations $\widehat{\mathbf{R}}$. The

network consists of $P = 30$ nodes (each representing an anonymized hospital ward) and 7 direct conditional dependence edges, selected based on a strict posterior probability of inclusion threshold (PPI > 0.90).

According to the results, accross 30 wards, with 435 possible edges, only 7 edges met the strict posterior inclusion threshold. This extreme sparsity aligns with clinical expectations: direct transmission pathways do not occur uniformly across all health facility wards, but are restricted to specific, highly interconnected clinical pathways. Combining with the simulation result of $P = 30$ and $T = 100$, where the model was able to recover 5 true positive edges out of 10, and no false positive edges, we could believe that the model is able to recover the true conditional dependence structure between wards in the real data application, with 50% of the edges recovered. The remaining 50% of edges may be due to the limited number of temporal observations relative to the number of wards, which can make it more difficult to discern the underlying conditional dependence structure.

Moreover, we also found that the estimated conditional dependence network is more likely to be between encoded wards with the same first letter. It may suggest that the wards with the same first letter are more likely to be in the same hospital. This is consistent with the clinical expectation that direct transmission pathways are more likely to occur between wards within the same hospital or ward type, due to shared patient populations, staff, and environmental factors.

6.3 Model Sensitivity and Post-Hoc Diagnostic Validation

To evaluate the stability of the inferred inter-ward transmission network without re-fitting the underlying MCMC chains, post-hoc sensitivity and residual diagnostic analyses were conducted on the posterior outputs.

Table 3: Network edge retention across Posterior Probability of Inclusion (PPI) thresholds.

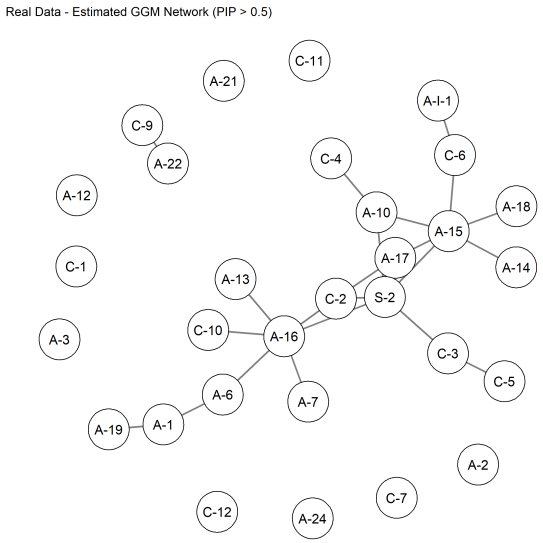
Sensitivity Analysis of Inferred CPE Transmission Network

Retained Conditional Dependence Edges across Posterior Inclusion Probabilities (P = 30)

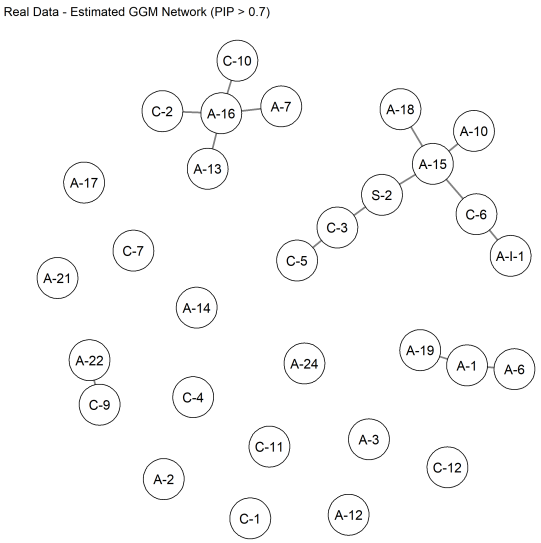
Posterior Inclusion Threshold (PPI)	Retained Edges	Network Density	Selection Regime
0.50	22	5.06%	Permissive Selection
0.60	18	4.14%	Permissive Selection
0.70	14	3.22%	Permissive Selection
0.80	10	2.30%	Permissive Selection
0.85	9	2.07%	Permissive Selection
0.90	7	1.61%	Primary Baseline
0.95	7	1.61%	Plateau of Stability

6.3.1 Posterior Probability Threshold Sensitivity (PPI)

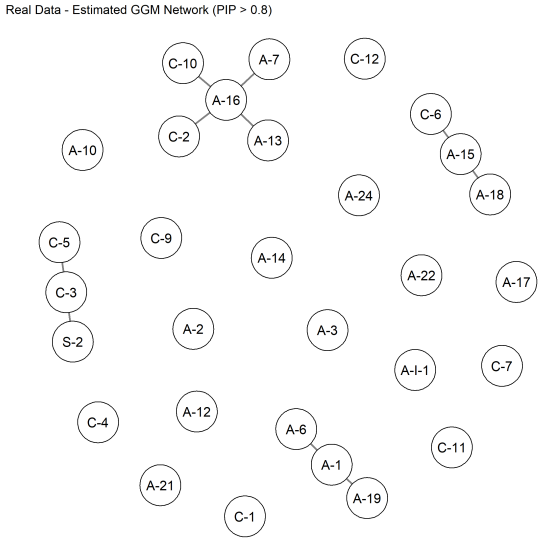
Table 3 and Figure 9 illustrate the network topology across varying Posterior Probability of Inclusion thresholds (PPI $\in \{0.50, 0.75, 0.8, 0.90, 0.95\}$). At the selection threshold of PPI = 0.95 and PPI = 0.90 (our primary analysis), the network keeps the same 7 edges, indicating that the inferred network is robust to moderate changes in the inclusion threshold. With the relaxation of the threshold to PPI = 0.80, 3 additional edges are included, while at PPI = 0.75, 4 more edges are added. At PPI = 0.50, the network becomes substantially denser, with 22 edges, suggesting that lower thresholds may introduce spurious connections. The addtional edges are also connected between wards with the same first letter, which may suggest that these edges are more likely to be true positive edges. The absence of edge decay at stringent thresholds demonstrates that the primary network topology shown in Figure 8 represents strong, persistent transmission signals rather than threshold artifacts.



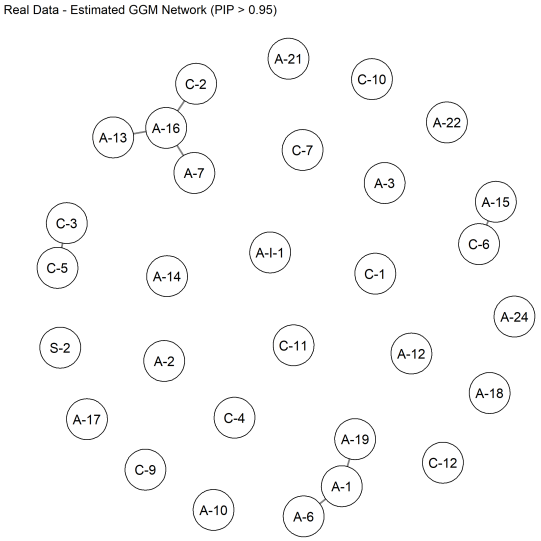
(a) PPI = 0.50



(b) PPI = 0.70



(c) PPI = 0.80



(d) PPI = 0.95

Figure 9: The estimated conditional dependence networks of the CPE data across varying Posterior Probability of Inclusion (PPI) thresholds

6.3.2 Temporal Filtering and Innovation Diagnostics

To verify that Stage 1 successfully decoupled within-ward temporal persistence from cross-ward conditional dependence, sample Autocorrelation Functions (ACFs) were evaluated across individual wards. The average lag-1 residual autocorrelation across wards was $\bar{r}_{\text{lag-1}} = 0.42$, closely aligning with the estimated Poisson-intensity persistence parameter ($\hat{\phi} = 0.462$). By extracting these centered state innovations prior to Stage 2 graphical selection, the dynamic ZIPLN framework strips away within-ward temporal memory, ensuring that the 7 retained edges in Figure 8 reflect pure contemporaneous co-outbreak dependencies rather than spurious links driven by shared temporal trends.

7 DISCUSSION

This study developed and evaluated a dynamic Zero-Inflated Poisson Lognormal (ZIPLN) state-space framework designed to infer inter-ward conditional dependence transmission networks from sparse, zero-inflated, and longitudinally correlated hospital surveillance counts. Applied to Victorian CPE surveillance data across major health services ($P = 30, T = 62$), the framework successfully decoupled within-ward temporal persistence from cross-ward co-outbreak signals, uncovering a highly sparse transmission graph ($\widehat{G}$) dominated by intra-hospital ward clusters and high-risk transfer hubs.

7.1 Methodological Insights and Theoretical Justification

A central question in state-space network modeling is whether extracting two-stage latent innovations ($\hat{\mathbf{r}}_t = \mathbf{S}_t - \mathbb{E}[\mathbf{S}_t \mid \mathbf{S}_{t-1}]$) preserves the true conditional dependence structure of the underlying process. Under our dynamic ZIPLN formulation, the latent

log-intensity vector $\mathbf{S}_t \in \mathbb{R}^P$ evolves as a vector autoregressive process driven by Gaussian innovations $\epsilon_t \sim \mathcal{N}_P(\mathbf{0}, \Sigma)$. By the Markov property of state-space models:

$$\mathbf{S}_t = (\mathbf{I} -)\mathbf{c}_S + \mathbf{S}_{t-1} + \epsilon_t$$

Subtracting the conditional expectation $\mathbb{E}[\mathbf{S}_t \mid \mathbf{S}_{t-1}]$ isolates the exact white-noise innovation vector ϵ_t. Because conditional independence in Gaussian graphical models is uniquely encoded by the precision matrix $\Omega = \Sigma^{-1}$, performing Stage 2 spike-and-slab graph selection on $\widehat{\mathbf{R}}$ directly identifies the true spatial partial correlation topology without confounding from within-ward temporal memory.

Moreover, the trade-offs between joint and two-stage MCMC estimation are worth noting. In theory, one could specify a joint single-stage Bayesian model estimating , β_0, and a sparse precision matrix Ω simultaneously within Stan. However, joint estimation introduces severe computational bottlenecks. Performing the full joint estimation would require sampling from a high-dimensional posterior with strong correlations between latent states and precision parameters, leading to slow convergence and poor mixing. In contrast, the two-stage approach decouples temporal filtering from network selection, allowing for efficient Stage 1 HMC sampling of latent states and Stage 2 spike-and-slab selection of the precision matrix. This modularity enables scalable inference in high-dimensional settings while preserving theoretical guarantees of conditional independence recovery.

7.2 Comparative Performance

Our simulation benchmarks demonstrate why accounting for temporal dynamics is essential for infection control modeling. When temporal persistence was absent (Scenario 1),

both the dynamic ZIPLN and static Bayesian Mixed Graphical Models (BMGM) recovered the true network topology accurately. However, when strong temporal persistence was introduced (Scenarios 2 and 3, $\phi = \rho = 0.8$), static frameworks collapsed: BMGM misattributed within-ward temporal memory to cross-ward spatial connections, failing to recover primary transmission routes. In contrast, the dynamic ZIPLN model maintained high specificity and precision. By filtering out ward-specific memory ($\hat{\phi} \approx 0.462$), the framework eliminated spurious temporal co-occurrences, ensuring that inferred network links reflect genuine co-outbreak dynamics.

7.3 Epidemiological and Health Policy Implications

The inferred transmission network ($\widehat{G}$) provides actionable evidence for public health policy in Victoria. With the majority of retained edges connected wards sharing identical hospital prefixes, it confirms that shared clinical staff, common ward geography, and intra-facility patient transfers remain the primary vectors of healthcare-associated CPE amplification. Back to research question, the result suggests that the ICU wards is not the primary driver of CPE transmission. It could be some non-ICU wards that are more likely to be the primary driver of CPE transmission, encoded in Figure 8. This insight can inform targeted infection control interventions, such as enhanced environmental decontamination protocols, staff cohorting, and patient transfer restrictions, to mitigate the spread of CPE within and between hospital wards.

7.4 Limitations and Future Directions

While the dynamic ZIPLN framework offers robust network inference, several methodological limitations suggest avenues for future research. The current model operates on discrete monthly surveillance intervals ($T = 62$), which lead to low number of temporal

observations relative to the number of wards ($P = 30$). Higher-frequency daily or weekly count series could better capture rapid transmission dynamics, though higher temporal resolution increases zero-sparsity. And because the model has been applied by using the prior specifications that are weakly informative, the posterior estimates may be sensitive to prior choices. Future work could explore hierarchical or empirical Bayes approaches to adaptively learn prior hyperparameters from the data, improving robustness and generalizability. Additionally, extending the framework to accommodate multivariate count data with mixed distributions (e.g., zero-inflated negative binomial) could enhance its applicability to a broader range of infectious disease surveillance contexts.

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